

EXCITATION-DEPENDENT IDENTIFIABILITY OF LATENT HIGHER-ORDER STRUCTURE FROM EDGE FLOWS

Weiping Yan

University of Minnesota, Twin Cities
Minneapolis, MN, USA

ABSTRACT

Which triangles of a network carry higher-order interactions? Edge flows reveal the latent filled-triangle set $\mathcal{S}$ only through their curl, and we show that *what* is identifiable is governed by the excitation covariance $\Gamma_{\mathcal{S}}$ of the triangle signals, through three regimes. For positive-diagonal $\Gamma_{\mathcal{S}}$ (known or unknown), the weighted support is identifiable at *any* rank deficiency of $\mathbf{B}_2$, because the lifted signatures $\{\mathbf{b}_\tau \mathbf{b}_\tau^\top\}$ are always linearly independent. For unrestricted PSD $\Gamma_{\mathcal{S}}$, the population curl covariance Σ_z determines only the achievable set $\mathcal{C}_{\mathcal{S}} = \{\mathbf{M} \succeq 0 : \text{im } \mathbf{M} \subseteq \text{im } \mathbf{U}_{\mathcal{S}}\}$, hence the support solely up to $\{\mathcal{S} : \text{im } \mathbf{M} \subseteq \text{im } \mathbf{U}_{\mathcal{S}}\}$; a strictly positive-diagonal but correlated $\Gamma_{\mathcal{S}}$ already defeats identification (a rank-one witness on K_4). At the projector excitation $\Gamma_{\mathcal{S}} = \sigma_c^2 (\mathbf{B}_{2,\mathcal{S}}^\top \mathbf{B}_{2,\mathcal{S}})^+$ the covariance collapses to $\sigma_n^2 \mathbf{I} + \sigma_c^2 \mathbf{P}_{\text{im}}$ exactly, so equal-image supports are indistinguishable at every SNR and sample size (noise level known throughout). This projector is exactly the Hodge-smoothness prior used in prior topology learning [1]. In the diagonal regime we give a lifted-covariance non-negative least-squares estimator that is consistent with a fully explicit $O(1/N)$ failure bound, validated on K_4 – K_8 against subspace and greedy baselines, including an excitation interpolation under which equal-image distinguishability provably vanishes. On real data, the same curl statistic *localizes* genuine unplanted rotational structure — tropical cyclones in ERA5 winds — against a same-field vorticity reference and the independent IB-TrACS archive. Code, data, and a 51-test suite numerically checking every constant are released.

Index Terms— Topological signal processing, simplicial complexes, identifiability, Hodge decomposition, covariance estimation, edge flows.

1. INTRODUCTION

Edge flows — currency exchange, traffic, ranking comparisons — carry higher-order structure on the triangles of a simplicial complex [2, 3, 4]. Topological signal processing builds filters and detectors on the Hodge Laplacian [5, 6, 7, 8, 9], and a recurring primitive is that the set $\mathcal{S}$ of “filled” triangles is unknown. One line *estimates* $\mathcal{S}$ algorithmically (MAP under a Hodge-smoothness prior [1], greedy learning [10], sparse cell-complex augmentation [11]); a second-order line models the edge covariance (PCA/total-variation on the sample covariance [2, §VII], ML fitting of a Hodge-structured simplicial Gaussian model [12]); a third *detects* signals in a *known* Hodge subspace [13]. Each fixes a model and gives an algorithm; to our knowledge none characterizes, as a function of the excitation class, *when* the filled-triangle structure is identifiable from the edge

covariance — the gap this paper fills: *what part of the latent structure is identifiable, and when?*

The answer is not a property of the geometry alone: it depends on the *excitation*. We model triangle signals as $\mathbf{y}_{\mathcal{S}} \sim \mathcal{N}(\mathbf{0}, \Gamma_{\mathcal{S}})$ with PSD excitation covariance $\Gamma_{\mathcal{S}}$ and show that the identifiable object moves with the regime of $\Gamma_{\mathcal{S}}$ (Theorem 1): from the full weighted support (positive diagonal, any rank deficiency), to only the achievable-covariance set $\mathcal{C}_{\mathcal{S}}$ — so the support merely up to $\{\mathcal{S} : \text{im } \mathbf{M} \subseteq \text{im } \mathbf{U}_{\mathcal{S}}\}$ — under unrestricted PSD excitation, to provable indistinguishability of equal-image supports at the projector excitation. The last case is not a corner curiosity: the Hodge-smoothness prior of [1], $\mathbf{y}_{\mathcal{S}} \sim \mathcal{N}(\mathbf{0}, \mathbf{L}_2^+)$, $\mathbf{L}_2 = \mathbf{B}_2^\top \mathbf{B}_2$, is this projector excitation — equal-image supports then have identical marginalized covariance likelihood, so any choice between them comes from the prior, not the data. We thus correct a folklore “rank obstruction” and delimit what covariance- and image-level methods [2, 12, 13] can resolve. Our tools are classical — detection theory and Fano arguments [14, 15], in the lineage of support-recovery and sparse-covariance limits [16, 17, 18, 19] — applied through a reduction (curl annihilation, after HodgeRank’s curl-as-inconsistency reading [20]) that makes them bite. Contributions:

1. *Theory*: the three excitation regimes (Thm. 1), with global analytic separation constants for the isotropic regime — equal-image binary supports are separated by at least 9 (attained by adding a tetrahedron’s fourth face) and equal-cardinality supports by 16 (swaps), proved for *every* clique complex via a Johnson-graph eigenvalue argument — and Chernoff exponents, for fixed graph and support as $\rho_2 \rightarrow 0$, whose worst case equals one isolated-triangle detection (Thm. 2).
2. *Achievability*: a lifted-covariance NNLS estimator for the diagonal regime, consistent with a fully explicit $O(1/N)$ failure bound in which every constant is derived and numerically checked (Thm. 3); Monte-Carlo validation on K_4 – K_8 with random supports against oracle-aided subspace and greedy baselines, and an excitation-interpolation experiment in which equal-image distinguishability collapses as predicted (Fig. 1).
3. *Real data, carefully positioned*: curl-based *vortex localization* — not topology recovery — of unplanted tropical-cyclone circulation in ERA5 winds, validated against a same-field vorticity reference and the independent IB-TrACS archive, with a classical baseline and moving-block bootstrap confidence intervals (Fig. 2).

2. MODEL AND CURL REDUCTION

Fix a graph $(\mathcal{V}, \mathcal{E})$ with node–edge incidence $\mathbf{B}_1$ and, for a candidate triangle set $\mathcal{T}$ (3-cliques), edge–triangle incidence $\mathbf{B}_2 \in \mathbb{R}^{|\mathcal{E}| \times p}$;

$B_1 B_2 = \mathbf{0}$, and each column $\mathbf{b}_\tau$ has three ± 1 entries, so $\mathbf{G} = B_2^\top B_2$ has $G_{\tau\tau} = 3$ and $G_{\sigma\tau} \in \{0, \pm 1\}$ (nonzero iff the triangles share an edge). We observe N i.i.d. snapshots

$$\mathbf{f}_t = B_1^\top \mathbf{a}_t + B_{2,S} \mathbf{y}_t + \mathbf{h}_t + \mathbf{n}_t, \quad t = 1, \dots, N, \quad (1)$$

with gradient nuisance $\mathbf{a}_t \sim \mathcal{N}(\mathbf{0}, \sigma_g^2 \mathbf{I})$, harmonic nuisance $\mathbf{h}_t$, noise $\mathbf{n}_t \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{I})$, and triangle excitation

$$\mathbf{y}_t \sim \mathcal{N}(\mathbf{0}, \Gamma_S), \quad \Gamma_S \succeq \mathbf{0}, \quad (2)$$

where $\Gamma_S \succeq \mathbf{0}$ is the excitation covariance; making it explicit is the point of this paper (prior generative models are special cases of it, §1). We take the harmonic nuisance to be *candidate-orthogonal*, $\mathbf{h}_t \in \ker B_2^\top$ (orthogonal to the curl space of the *full* candidate complex); the gradient part is annihilated unconditionally since $B_2^\top B_1^\top = \mathbf{0}$. Then curl projection removes both nuisances exactly [20]: in an orthonormal basis $\mathbf{Q}$ of $\text{im } B_2$ ($r = \text{rank } B_2$, B_2 the full candidate incidence), the sufficient statistic $\mathbf{z}_t = \mathbf{Q}^\top \mathbf{f}_t$ is Gaussian with

$$\Sigma_z(S, \Gamma_S) = \sigma_n^2 \mathbf{I}_r + U_S \Gamma_S U_S^\top, \quad U = \mathbf{Q}^\top B_2, \quad (3)$$

with $\|\mathbf{u}_\tau\|^2 = 3$ and $\mathbf{u}_\sigma^\top \mathbf{u}_\tau = G_{\sigma\tau}$. Identifiability of S from (3) is the object of study; for the isolated-triangle scalar test this reduces to the classical two-variance problem with curl-SNR $\rho = 3\sigma_c^2/\sigma_n^2$ [14] (supplement §S1–S3 develops that line: detection thresholds, Fano converses, decorrelated detectors, plug-in estimation).

Definition 1 (Observational equivalence and identifiability). *All statements here are about the population Σ_z (finite- N estimation is §4). Two pairs (S, Γ_S) and $(S', \Gamma_{S'})$ are observationally equivalent if $\Sigma_z(S, \Gamma_S) = \Sigma_z(S', \Gamma_{S'})$; a functional of (S, Γ_S) is identifiable on an excitation class $\mathcal{G}$ if it is constant on each equivalence class with excitations in $\mathcal{G}$. The realized range is $\mathcal{R}(S, \Gamma_S) = \text{im}(\Sigma_z - \sigma_n^2 \mathbf{I}) = \text{im}(U_S \Gamma_S^{1/2}) \subseteq \text{im } U_S$, always identifiable since it is a function of Σ_z alone. If $\Gamma_S \succ \mathbf{0}$ then $\mathcal{R} = \text{im } U_S$; when $B_{2,S}$ has full column rank (so U_S is injective) the converse holds as well.*

3. EXCITATION-DEPENDENT IDENTIFIABILITY

Lemma 1 (Lifted spark). *For any candidate set of distinct 3-cliques of a simple graph, the matrices $\{\mathbf{b}_\tau \mathbf{b}_\tau^\top\}_{\tau \in \mathcal{T}}$ are linearly independent.*

Proof. A pair of distinct edges lies in at most one common triangle, so for $e \neq e'$ both in τ , the (e, e') entry of $\sum_\sigma w_\sigma \mathbf{b}_\sigma \mathbf{b}_\sigma^\top$ equals $\pm w_\tau$: every coefficient is read off a distinct matrix entry. $\square$

Theorem 1 (Three excitation regimes). *Let σ_n^2 be known and write $\mathbf{M} = \mathbf{M}(S, \Gamma) = U_S \Gamma U_S^\top = \Sigma_z - \sigma_n^2 \mathbf{I}$ in (3). The regimes below are overlapping constraints on Γ_S , not a partition (e.g. $\sigma_c^2 \mathbf{I}$ is both diagonal (a) and a special full-rank PSD (b)). (a) Positive diagonal. If $\Gamma_S = \text{diag}(\gamma_\tau)_{\tau \in S}$ with $\gamma_\tau > 0$ (known or unknown), then $\mathbf{M} = \sum_{\tau \in S} \gamma_\tau \mathbf{u}_\tau \mathbf{u}_\tau^\top$ and, by Lemma 1, the map $(S, \Gamma_S) \mapsto \Sigma_z$ is injective: the weighted support is identifiable at any rank deficiency of B_2 , including all $\binom{n}{3}$ triangles of K_n . (b) Arbitrary PSD. The achievable set $\mathcal{C}_S = \{U_S \Gamma U_S^\top : \Gamma \succeq \mathbf{0}\} = \{\mathbf{M} \succeq \mathbf{0} : \text{im } \mathbf{M} \subseteq \text{im } U_S\}$, so the population covariance determines $\mathbf{M}$ in full but pins the support only to $\{S : \text{im } \mathbf{M} \subseteq \text{im } U_S\}$: the realized range $\mathcal{R} = \text{im } \mathbf{M} \subseteq \text{im } U_S$ is always readable, yet the candidate image $\text{im } B_{2,S}$ is not (a larger-image S' with a suitable Γ gives the same*

$\mathbf{M}$; $\Gamma_S \succ \mathbf{0}$ does recover $\mathcal{R} = \text{im } U_S$). The obstruction is correlation, not zero variance: on K_4 (all four faces) with $\mathbf{c} \in \ker U_S$ and $\mathbf{v} = \mathbf{e}_1 - \frac{1}{4}\mathbf{c} = (.75, .25, -.25, .25)^\top$, the strictly-positive-diagonal $\Gamma' = \mathbf{v}\mathbf{v}^\top$ gives $U_S \Gamma' U_S^\top = \mathbf{u}_1 \mathbf{u}_1^\top$ — identical to the singleton $\{\tau_1\}$ though the images are 3- vs. 1-dimensional. Regime (a) additionally requires Γ diagonal. (c) Projector excitation. For $\Gamma_S = \sigma_c^2 (B_{2,S}^\top B_{2,S})^+$, $\mathbf{M} = \sigma_c^2 \mathbf{P}_{\text{im } U_S}$ exactly; supports with equal curl image induce identical snapshot distributions and are indistinguishable at every SNR and every N .

Proof sketch. (a) Read the weights off the lifted representation (spark). (b) $\subseteq$ is immediate; $\supseteq$: given PSD $\mathbf{M}$ with $\text{im } \mathbf{M} \subseteq \text{im } U_S$, take $\Gamma = U_S^\top \mathbf{M} (U_S^\top)^\top \succeq \mathbf{0}$. (c) $\mathbf{A}(\mathbf{A}^\top \mathbf{A})^+ \mathbf{A}^\top = \mathbf{P}_{\text{im } \mathbf{A}}$ (SVD). On K_n , $\dim \text{im } B_2 = \binom{n-1}{2}$ (a $3/n$ DoF ratio), and equal-image supports exist whenever a tetrahedron is hosted ($\mathbf{b}_{012} - \mathbf{b}_{013} + \mathbf{b}_{023} - \mathbf{b}_{123} = \mathbf{0}$). Full proofs and numerical certificates: supplement §S4, tests/test_excitation.py. $\square$

Remark 1 (Scope). (i) *Deterministic unknown signals are the rank-one degenerate boundary of regime (b): the realized range is a single direction, so not even the image is seen.* (ii) *On K_n with σ_n unknown, regime (a) retains exactly one ambiguous direction, since $\sum_{\tau \in \mathcal{T}} \mathbf{u}_\tau \mathbf{u}_\tau^\top = n \mathbf{I}_r$: a uniform weight shift trades off against the noise floor.* (iii) *Regime (b) delimits image-level methods — e.g. matched subspace detectors [13] — whose population statistics see only the realized range $\text{im } \mathbf{M} \subseteq \text{im } B_{2,S}$: without excitation structure, no method can do better.*

Theorem 2 (The price in the isotropic class). *Let $\Gamma_S = \sigma_c^2 \mathbf{I}$, $\rho_2 = \sigma_c^2/\sigma_n^2$, and write the dimensionless signature Gram $\mathbf{D}_S = \sum_{\tau \in S} \mathbf{u}_\tau \mathbf{u}_\tau^\top$ (so $\mathbf{M}_S = \sigma_c^2 \mathbf{D}_S$). For any clique complex: distinct binary supports with equal curl image satisfy $\|\mathbf{D}_S - \mathbf{D}_{S'}\|_F^2 \geq 9$, with equality exactly for single-face differences (e.g. $S' = S \cup \{\text{fourth face}\}$), and ≥ 16 under equal cardinality (tetrahedral swaps). For fixed graph and support pair, as $\rho_2 \rightarrow 0$ the optimal-test Chernoff exponents are $C_G = \rho_2^2 \|\Delta \mathbf{D}\|_F^2 / 16 (1 + o(1))$, i.e. $\frac{9}{16} \rho_2^2 (1 + o(1))$ (subset) and $\rho_2^2 (1 + o(1))$ (swap) — and $\frac{9}{16} \rho_2^2$ equals the isolated-triangle detection exponent $C(\rho) = \rho^2 / 16$ at $\rho = 3\rho_2$: the hardest equal-image decision costs one isolated-triangle detection. Hence in this scope (fixed pair, isotropic excitation, leading low-SNR exponent) rank deficiency is free at the exponent level, and only the Fano log-multiplicity of the $4\binom{n}{4}$ tetrad hypotheses ($\asymp \log n$ on K_n) reflects the geometry.*

Proof sketch. Separations: $\|\sum_\tau c_\tau \mathbf{b}_\tau \mathbf{b}_\tau^\top\|_F^2 = \mathbf{c}^\top (9\mathbf{I} + \mathbf{A}) \mathbf{c}$ for ternary $\mathbf{c}$, where $\mathbf{A}$ is the share-an-edge adjacency — an induced subgraph of the Johnson graph $J(n, 3)$, so $\lambda_{\min}(\mathbf{A}) \geq -3$ by interlacing and $\text{sep} \geq 6\|\mathbf{c}\|^2$; enumerate $\|\mathbf{c}\|^2 \in \{1, 2\}$. Exponents: Bhattacharyya expansion gives $C_G = \rho_2^2 \|\Delta \mathbf{D}\|_F^2 / 16 (1 + o(1)) = \|\Delta \mathbf{M}\|_F^2 / (16\sigma_n^4) (1 + o(1))$ (the two forms coincide). Numerically checked: swap C_G/ρ_2^2 within 6×10^{-4} of 1 and subset matching the single-triangle exponent to 3×10^{-5} relative (at $\rho_2 = 10^{-4}, 10^{-5}$); separations exhaustive over all 2^{10} supports of K_5 . Proofs: supplement §S4. $\square$

Proposition 1 (Edge-sharing marginal laws). *Assume B_2 full column rank and $\Gamma_S = \sigma_c^2 \mathbf{I}$. The GLS/BBLUE inversion $\hat{\mathbf{y}}_t = \mathbf{G}^{-1} B_2^\top \mathbf{f}_t$ [21] satisfies, exactly, $\hat{\mathbf{y}}_t = \tilde{\mathbf{y}}_t + \mathbf{e}_t$ with $\mathbf{e}_t \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{G}^{-1})$: each coordinate is a two-variance test with effective SNR $\rho_{\tau}^{\text{eff}} = \sigma_c^2 / (\sigma_n^2 (\mathbf{G}^{-1})_{\tau\tau})$ — the classical variance-inflation factor as the price of edge-sharing — and exact recovery admits a correlation-robust union-bound guarantee. The noise coordinates stay correlated ($\sigma_n^2 \mathbf{G}^{-1}$ is not diagonal), so the joint law does not*

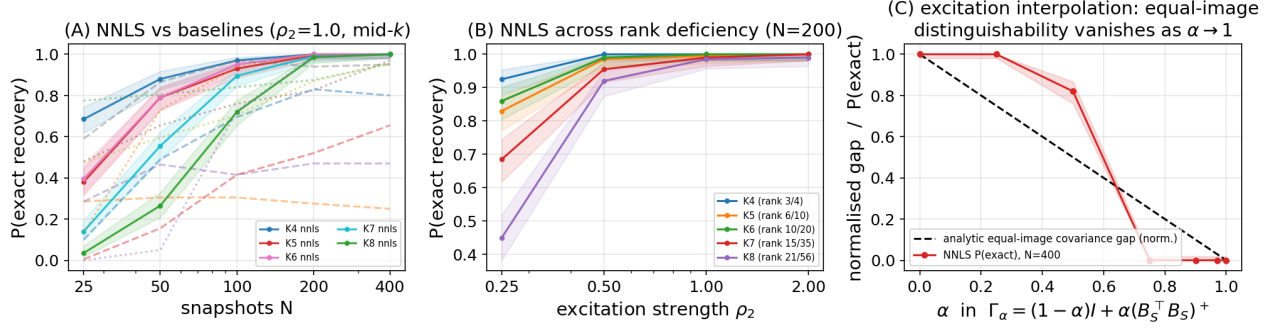


Fig. 1. Second-order achievability. (A) Exact recovery vs. N on K_4 – K_8 (random supports, $\rho_2 = 1$): NNLS (solid, 95% Wilson bands) vs. oracle-aided subspace (dashed) and greedy (dotted) baselines. (B) NNLS vs. ρ_2 at $N = 200$ across rank deficiency (legend: rank $\mathbf{B}_2/p$). (C) Excitation interpolation on the K_5 tetrad: the analytic equal-image gap (normalized) closes only at $\alpha = 1$ (Thm. 1(c)); thresholded NNLS recovery collapses earlier (zero from $\alpha = 0.75$).

factorize; benchmarks, the naive-detector failure mode, and proofs are in supplement §S1, S3.

Proposition 2 (Detection and joint-recovery scalings). *In the isotropic class, deciding one isolated triangle is the two-variance Gaussian test with variances $3\sigma_n^2$ vs. $9\sigma_c^2 + 3\sigma_n^2$ [14]: attaining error $\leq \delta$ at the exponent level requires the curl-SNR scale $\rho \geq \rho^*(N) = \Theta(\sqrt{\log(1/\delta)/N})$ ($C(\rho) = \rho^2/16 + o(\rho^2)$ [15]). Recovering a uniformly random size- k support among p edge-disjoint candidates with error $\leq \epsilon$ requires $N \geq [(1 - \epsilon) \log \binom{p}{k} - \log 2]/(k \text{KL}_1)$, $\text{KL}_1 = (\rho - \log(1 + \rho))/2$ (Fano): joint recovery pays a $\log \binom{p}{k}$ factor. Signal-agnostic and sub-Gaussian variants, the exact edge-disjoint product law, partial edge observation (q^{3k} laws), and plug-in calibration are developed in supplement §S1–S3; these scalings are classical [16] — what is problem-specific is the reduction that produces ρ and the constants (3, 9).*

4. ACHIEVABILITY: LIFTED-COVARIANCE NNLS

In class (a) the theory says the weighted support is identifiable; the matching estimator is non-negative least squares on the lifted dictionary $\mathbf{A} = [\text{vec}(\mathbf{u}_\tau \mathbf{u}_\tau^\top)]_\tau$:

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \geq 0} \left\| \hat{\Sigma}_z - \sigma_n^2 \mathbf{I} - \sum_\tau w_\tau \mathbf{u}_\tau \mathbf{u}_\tau^\top \right\|_F^2, \quad (4)$$

with $\hat{\Sigma}_z = \frac{1}{N} \sum_t \mathbf{z}_t \mathbf{z}_t^\top$, $\hat{S} = \{\tau : \hat{w}_\tau > w_{\min}/2\}$, and $w_{\min} = \min_{\tau \in S} w_\tau$.

Theorem 3 (Consistency and explicit failure bound). *$\hat{\mathbf{w}} \rightarrow \mathbf{w}$ almost surely, and for every N ,*

$$\Pr[\hat{S} \neq S] \leq \frac{16[(\text{tr } \Sigma_z)^2 + \|\Sigma_z\|_F^2]}{N \sigma_{\min}^2(\mathbf{A}) w_{\min}^2}. \quad (5)$$

Proof sketch. Three derived steps, each numerically checked: (i) cone-constrained least squares obeys the deterministic bound $\|\hat{\mathbf{w}} - \mathbf{w}\|_2 \leq 2\|\hat{\Sigma}_z - \Sigma_z\|_F / \sigma_{\min}(\mathbf{A})$ — optimality of $\hat{\mathbf{w}}$, feasibility of $\mathbf{w}$, Cauchy–Schwarz — with $\sigma_{\min}(\mathbf{A}) > 0$ by Lemma 1; (ii) for Gaussian snapshots $\mathbb{E}\|\hat{\Sigma}_z - \Sigma_z\|_F^2 = [(\text{tr } \Sigma_z)^2 + \|\Sigma_z\|_F^2]/N$ exactly (Wishart second moments); (iii) Markov at threshold $w_{\min}/2$. The bound is conservative (Markov) — its value is the explicit $O(1/N)$ rate with computable constants; empirically the transition occurs roughly an order of magnitude earlier (Fig. 1). Proof: supplement §S4. $\square$

Fig. 1 validates the estimator across K_4 – K_8 with per-trial random supports at several sparsities k , a full (N, ρ_2) grid with $N \in \{25, \dots, 400\}$ and $\rho_2 \in \{0.25, \dots, 2\}$, 200 Monte-Carlo trials per cell, and 95% Wilson intervals (`run_second_order.py`; every cell, including the derived bound (5) evaluated at that cell, is in the released JSON). Protocol: the headline threshold is the theorem’s $w_{\min}/2$ (planted weights known); an oracle-free largest-gap rule is logged alongside but is worse (grid-mean 0.63 vs. 0.69, ahead in 18/260 cells). The full edge-domain pipeline is used; the greedy comparator is a generic covariance-atom heuristic, and both baselines get oracle side information (subspace: true k and $\dim \text{im } \mathbf{B}_{2,S}$; greedy: true σ_n).

(A) Among these three implementations (not tuned literature methods), NNLS is the only one with exact recovery in every cell — incl. maximal rank deficiency (rank $\mathbf{B}_2/p = 21/56$ on K_8) — reaching 1 at mid- k by $N=400$, $\rho_2=1$ (0.99 at the densest K_8 cell, $k=28$). The oracle- k subspace baseline can lead at small N but collapses on dependent candidates (0.00 at K_8 , $k=28$): its population scores tie there (Remark 1(iii)), and its finite- N tie-breaking rides on sample eigen-anisotropy, a side channel the projector excitation removes entirely. Greedy stalls at sparse supports (0.43 on K_5 at $k=2$, $N=400$). (B) Recovery improves monotonically in ρ_2 uniformly across rank deficiency — the empirical face of Theorem 1(a). The bound (5) upper-bounds every cell; being a Markov bound it is conservative by one to three orders of magnitude (the empirical failure decays exponentially, the bound as $1/N$). It is non-vacuous (clipped value < 1) at only 7 of the 260 grid cells — the small- $w_{\min}$, small- N corner, where it falls as low as 0.42 (and to ~ 0.75 at dedicated test cells) with zero observed failures; elsewhere the $O(1/N)$ rate is the guarantee. (C) Under $\Gamma_\alpha = \rho_2[(1 - \alpha)\mathbf{I} + \alpha(\mathbf{B}_{2,S}^\top \mathbf{B}_{2,S})^+]$ on the K_5 tetrad, with the same $w_{\min}/2$ rule (the diagonal model is deliberately misspecified for $\alpha > 0$ — that is the point), the analytic equal-image gap shrinks continuously and the empirical NNLS recovery of the specific S collapses before the gap reaches zero — thresholded estimation dies earlier than distinguishability — with the threshold-free endpoint at $\alpha = 1$ on Theorem 1(c): the $m=4$ equal-image supports have identical distributions, so no estimator beats chance $1/m$; the plotted 0 is specific- S recovery, itself $\leq 1/m$. Identifiability is a property of the excitation, not the geometry alone.

5. REAL DATA: CURL-BASED VORTEX LOCALIZATION

We position the real-data study precisely: it is *vortex localization*, not filled-triangle topology recovery — the mesh below is

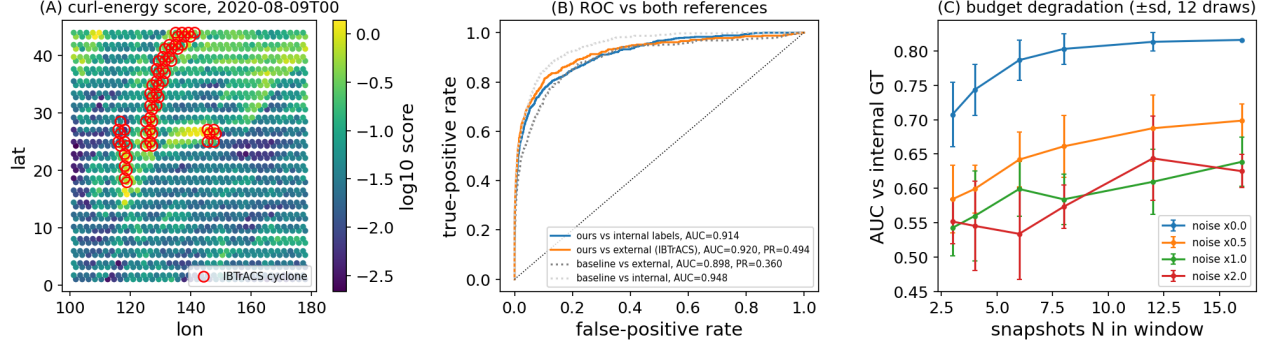


Fig. 2. Curl-based vortex localization on real data. (A) Centered curl-energy scores (vorticity scale) for one 4-day window; red circles: independent IBTrACS fixes. (B) ROC pooled over all windows: same-field vorticity reference, independent IBTrACS labels (with PR-AUC), and the classical coarse-vorticity baseline. (C) Budget degradation (mean $\pm$ sd over 12 subsample draws; uniform centered statistic, $N \geq 3$).

simply connected, hence full-column-rank with no equal-image confusers, and real weather carries no latent boolean support. What it demonstrates is that the paper’s sufficient statistic localizes *genuine, unplanted* rotational structure on a real vector field. Construction: ERA5 10m winds [22] over the Western North Pacific (Aug–Sep 2020, $0.7^\circ/6$ -hourly) are subsampled to a 2.1° right-triangulated mesh (836 nodes, 2389 edges, 1554 triangles; $\#\text{triangles} = |\mathcal{E}| - |\mathcal{V}| + 1$ by Euler, so $\mathbf{B}_2$ is full column rank); edge flows are trapezoidal line integrals of the wind in a local equirectangular metric. By Stokes’ theorem a triangle’s curl is its circulation, so the temporally centered curl-energy score (area²-normalized to the mean-vorticity scale; no oracle parameters) ranks cyclone-bearing triangles within 4-day windows of 16 snapshots, pooled over the season’s 15 windows for evaluation. Storms translate several triangle-widths per 4-day window; centering removes the window-mean flow, so the statistic responds to the time-varying circulation a moving vortex deposits. Validation (Fig. 2B) against a *same-field consistency reference* (finite-difference vorticity on the $3\times$ -finer source grid) gives AUC 0.914 (moving-block bootstrap 95% CI [0.894, 0.936]), and against the *genuinely independent* IBTrACS archive [23] — 13 storms, 726 fixes — AUC 0.920 [0.879, 0.951], PR-AUC 0.494 [0.273, 0.666] at prevalence 1.8%, and $P@k = 0.483$. Against a classical pointwise-vorticity baseline at the same information budget (mesh-node winds), the edge-flow statistic is better on *all three* independent external metrics (baseline: AUC 0.898, PR-AUC 0.360, $P@k = 0.389$) and slightly lower on the internal reference (0.914 vs. 0.948), which shares the baseline’s functional. The decorrelated (GLS) variant of Prop. 1 ranks worse here ($\mathbf{G}^{-1}$ amplifies noise on a near-regular mesh) and is reported in the repository. FX consistency checks and planted road-network studies from earlier revisions are retained in the repository and supplement, not claimed as recovery evidence.

Limitations. The i.i.d.-Gaussian snapshot model is an idealization of windowed weather; evaluation is therefore threshold-free ranking rather than support recovery, moving-block bootstrap intervals quantify pooling uncertainty, and the internal reference is a same-field consistency check — only IBTrACS is independent. The vorticity-threshold sensitivity of the internal AUC (0.876–0.959 over 10^{-5} – $5 \times 10^{-5} \text{ s}^{-1}$) is reported in the released JSON. Nothing here claims a new cyclone detector; the experiment grounds the paper’s sufficient statistic on genuinely unplanted structure.

6. CONCLUSION

Identifiability of latent higher-order structure from edge flows is excitation-dependent: fully identifiable under diagonal excitation at any rank deficiency; under unrestricted PSD excitation the population covariance pins down only the achievable-covariance set $\mathcal{C}_S$, so the support is determined merely up to $\{S : \text{im } \mathbf{M} \subseteq \text{im } \mathbf{U}_S\}$ (not even the candidate image in general); and provably ambiguous at the projector excitation — with global analytic separation constants, a worst-case exponent equal to one isolated-triangle detection, and a consistent NNLS estimator carrying a fully explicit $O(1/N)$ bound. The three regimes delimit both algorithmic structure learning [1, 10, 11] and subspace detection [13]; on real data the same statistic localizes unplanted cyclone circulation. Open problems: identifiability *between* the diagonal and free-PSD excitation classes (e.g. banded or Toeplitz $\mathbf{\Gamma}_S$ — where between reading off every weight and seeing only the image does the transition occur?); sharp first-order sparse thresholds over $\ker \mathbf{B}_2$; effective sample sizes N_{eff} under temporal dependence; and tightening the Markov factor in (5) to match the empirical transition.

7. REFERENCES

- [1] Sravanthi Gurugubelli and Sundeep Prabhakar Chepuri, “Simplicial complex learning from edge flows via sparse clique sampling,” in *European Signal Processing Conference (EUSIPCO)*, 2024.
- [2] Sergio Barbarossa and Stefania Sardellitti, “Topological signal processing over simplicial complexes,” *IEEE Transactions on Signal Processing*, vol. 68, pp. 2992–3007, 2020.
- [3] Michael T. Schaub, Yu Zhu, Jean-Baptiste Seby, T. Mitchell Roddenberry, and Santiago Segarra, “Signal processing on higher-order networks: Livin’ on the edge... and beyond,” *Signal Processing*, vol. 187, pp. 108149, 2021.
- [4] Lek-Heng Lim, “Hodge Laplacians on graphs,” *SIAM Review*, vol. 62, no. 3, pp. 685–715, 2020.
- [5] Maosheng Yang, Elvin Isufi, Michael T. Schaub, and Geert Leus, “Simplicial convolutional filters,” *IEEE Transactions on Signal Processing*, vol. 70, pp. 4633–4648, 2022.
- [6] Claudio Battiloro, Paolo Di Lorenzo, and Sergio Barbarossa, “Topological Slepians: Maximally localized representations of signals over simplicial complexes,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.
- [7] Lucille Calmon, Michael T. Schaub, and Ginestra Bianconi, “Dirac signal processing of higher-order topological signals,” *New Journal of Physics*, vol. 25, pp. 093013, 2023.
- [8] Lorenzo Marinucci, Claudio Battiloro, and Paolo Di Lorenzo, “Topological adaptive least mean squares algorithms over simplicial complexes,” *IEEE Transactions on Signal and Information Processing over Networks*, vol. 12, pp. 283–298, 2026, arXiv:2505.23160.
- [9] Chengen Liu, Geert Leus, and Elvin Isufi, “Unrolling of simplicial Elasticnet for edge flow signal reconstruction,” *IEEE Open Journal of Signal Processing*, vol. 5, 2024, DOI 10.1109/OJSP.2023.3339376.
- [10] Andrei Buciulea, Elvin Isufi, Geert Leus, and Antonio G. Marques, “Learning the topology of a simplicial complex using simplicial signals: A greedy approach,” arXiv:2502.20159, 2025.
- [11] Josef Hoppe and Michael T. Schaub, “Representing edge flows on graphs via sparse cell complexes,” arXiv:2309.01632; Learning on Graphs Conference (LoG), 2023.
- [12] Lorenzo Marinucci, Gabriele D’Acunto, Paolo Di Lorenzo, and Sergio Barbarossa, “Simplicial Gaussian models: Representation and inference,” arXiv:2510.12983, 2025.
- [13] Chengen Liu, Victor M. Tenorio, Antonio G. Marques, and Elvin Isufi, “Matched topological subspace detector,” arXiv:2504.05892, 2025.
- [14] H. Vincent Poor, *An Introduction to Signal Detection and Estimation*, Springer, 2nd edition, 1994.
- [15] Thomas M. Cover and Joy A. Thomas, *Elements of Information Theory*, Wiley, 2nd edition, 2006.
- [16] Martin J. Wainwright, “Information-theoretic limits on sparsity recovery in the high-dimensional and noisy setting,” *IEEE Transactions on Information Theory*, vol. 55, no. 12, pp. 5728–5741, 2009.
- [17] Arash A. Amini and Martin J. Wainwright, “High-dimensional analysis of semidefinite relaxations for sparse principal components,” *The Annals of Statistics*, vol. 37, no. 5B, pp. 2877–2921, 2009.
- [18] Quentin Berthet and Philippe Rigollet, “Optimal detection of sparse principal components in high dimension,” *The Annals of Statistics*, vol. 41, no. 4, pp. 1780–1815, 2013.
- [19] T. Tony Cai, Cun-Hui Zhang, and Harrison H. Zhou, “Optimal rates of convergence for covariance matrix estimation,” *The Annals of Statistics*, vol. 38, no. 4, pp. 2118–2144, 2010.
- [20] Xiaoye Jiang, Lek-Heng Lim, Yuan Yao, and Yinyu Ye, “Statistical ranking and combinatorial Hodge theory,” *Mathematical Programming*, vol. 127, no. 1, pp. 203–244, 2011.
- [21] Steven M. Kay, *Fundamentals of Statistical Signal Processing: Estimation Theory*, Prentice-Hall, 1993.
- [22] Hans Hersbach, Bill Bell, Paul Berrisford, et al., “The ERA5 global reanalysis,” *Quarterly Journal of the Royal Meteorological Society*, vol. 146, no. 730, pp. 1999–2049, 2020.
- [23] Kenneth R. Knapp, Michael C. Kruk, David H. Levinson, Howard J. Diamond, and Charles J. Neumann, “The International Best Track Archive for Climate Stewardship (IBTrACS): Unifying tropical cyclone data,” *Bulletin of the American Meteorological Society*, vol. 91, no. 3, pp. 363–376, 2010.

Compliance with ethical standards

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Research involving human participants and/or animals. This article does not contain any studies with human participants or animals. All data used are public: ERA5 reanalysis (Copernicus/ECMWF, via the ARCO-ERA5 mirror), IBTrACS v04r01 (NOAA NCEI), TNTP road networks, and ECB reference exchange rates.

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